Reactions Unit: Sample Test

Name: _____ Pd: _____

You may use the solubility rules and the activity series provided. Part I is worth 1 point per question (20 points). Part II is on the following pages.

Part I: Multiple Choice

1. In the equation $4Fe + 3O_2 \rightarrow 2Fe_2O_3$, what does the subscript 3 in Fe_2O_3 represent?

A. Each formula unit of Fe_2O_3 contains 3 oxygen atoms.
B. 3 molecules of Fe_2O_3 are produced.
C. 3 moles of O_2 react.
D. The charge on the oxygen ion is 3.

2. How many chlorine atoms are on each side of the equation $2Fe + 3Cl_2 \rightarrow 2FeCl_3$?

A. 3 on the left and 3 on the right
B. 6 on the left and 3 on the right
C. 6 on the left and 6 on the right
D. 2 on the left and 6 on the right

3. Which equation is correctly balanced?

A. $Mg + O_2 \rightarrow MgO$
B. $2Mg + O_2 \rightarrow 2MgO$
C. $Mg + O_2 \rightarrow MgO_2$
D. $2Mg + 2O_2 \rightarrow 2MgO$

4. Which equation correctly shows the balanced decomposition of calcium chloride?

A. $CaCl_2 \rightarrow Ca + 2Cl$
B. $CaCl \rightarrow Ca + Cl$
C. $2CaCl_2 \rightarrow 2Ca + Cl_2$
D. $CaCl_2 \rightarrow Ca + Cl_2$

5. Which equation represents a combustion reaction?

A. $2Na + Cl_2 \rightarrow 2NaCl$
B. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
C. $2KClO_3 \rightarrow 2KCl + 3O_2$
D. $CaO + H_2O \rightarrow Ca(OH)_2$

6. What type of reaction is $Mg + 2HCl \rightarrow MgCl_2 + H_2$?

A. Single replacement
B. Synthesis
C. Double replacement
D. Decomposition

7. Which best classifies the reaction $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$?

A. Combustion
B. Single replacement
C. Decomposition
D. Neutralization

8. What are the products when water decomposes? $H_2O(l) \rightarrow$?

A. H and O
B. H_2O_2
C. H_2 and O_2
D. H_2 and O

9. What is the product when aluminum reacts with sulfur, $Al + S_8$?

A. AlS
B. Al_2S_3
C. Al_3S_2
D. Al_2S

10. Which substance must be a reactant in every combustion reaction?

A. CO_2
B. H_2O
C. N_2
D. O_2

11. Using the activity series provided, which reaction will NOT occur?

A. $Ag(s) + MgCl_2(aq) \rightarrow$
B. $Mg(s) + FeCl_2(aq) \rightarrow$
C. $Zn(s) + Pb(NO_3)_2(aq) \rightarrow$
D. $Cu(s) + AgNO_3(aq) \rightarrow$

12. Using the activity series provided, which halogen reaction will occur?

A. $I_2(s) + NaBr(aq) \rightarrow$
B. $Br_2(l) + NaCl(aq) \rightarrow$
C. $Cl_2(g) + NaI(aq) \rightarrow$
D. $I_2(s) + KCl(aq) \rightarrow$

13. What are the products when solid sodium reacts with liquid water?

A. Na_2O and H_2
B. NaH and O_2
C. NaOH and O_2
D. NaOH and H_2

14. Aqueous $Al_2(SO_4)_3$ is mixed with aqueous Na_3PO_4. What are the formulas of the products?

A. $AlPO_4$ and Na_2SO_4
B. $Al_3(PO_4)_2$ and Na_3SO_4
C. $AlPO_4$ and Na_3SO_4
D. Al_2PO_4 and $NaSO_4$

15. Aqueous $Pb(NO_3)_2$ is mixed with aqueous KI. Using the solubility rules provided, what is the precipitate?

A. KNO_3
B. PbI_2
C. PbI
D. K_2I

16. Using the solubility rules provided, which pair of aqueous solutions will produce NR (no reaction)?

A. $NaOH + FeCl_3$
B. $Na_2S + CuCl_2$
C. $LiBr + KNO_3$
D. $NaOH + Ni(NO_3)_2$

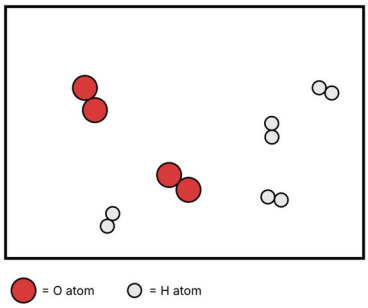
17. Aqueous K_2CO_3 is mixed with aqueous $CaCl_2$ and a white precipitate forms. Which is the net ionic equation?

A. $Ca^{2+}(aq) + CO_3^{2-}(aq) \rightarrow CaCO_3(s)$
B. $2K^+(aq) + 2Cl^-(aq) \rightarrow 2KCl(aq)$
C. $K_2CO_3(aq) + CaCl_2(aq) \rightarrow CaCO_3(s) + 2KCl(aq)$
D. $2K^+(aq) + CO_3^{2-}(aq) + Ca^{2+}(aq) + 2Cl^-(aq) \rightarrow CaCO_3(s) + 2K^+(aq) + 2Cl^-(aq)$

18. When writing a total ionic equation for a reaction in water, which species should NOT be written as separate ions?

A. $Na_2SO_4(aq)$
B. $KNO_3(aq)$
C. $Ba(NO_3)_2(aq)$
D. AgCl(s)

19. Hydrogen and oxygen react to form water: $2H_2 + O_2 \rightarrow 2H_2O$. The box below shows the mixture before the reaction. If the reaction goes to completion, how many H_2O molecules form?



A. 2
B. 4
C. 6
D. 8

20. What is the correct formula for iron(III) sulfate?

A. $FeSO_4$
B. $Fe_3(SO_4)_2$
C. $Fe_2(SO_4)_3$
D. Fe_2SO_4

Part II: Free Response

1) Classify each reaction as Synthesis (S), Decomposition (D), Combustion (C), Single Replacement (SR), Double Replacement (DR), or Neutralization (N), then balance it. *(12 pts)*

a. ___Mg + ___Fe_2O_3 → ___MgO + ___Fe Type: _____

b. ___Al + ___O_2 → ___Al_2O_3 Type: _____

c. ___C_8H_{18} + ___O_2 → ___CO_2 + ___H_2O Type: _____

d. ___$Fe_2(SO_4)_3$ + ___KOH → ___$Fe(OH)_3$ + ___K_2SO_4 Type: _____

e. ___Al_2O_3 → ___Al + ___O_2 Type: _____

f. ___H_3PO_4 + ___$Ca(OH)_2$ → ___$Ca_3(PO_4)_2$ + ___H_2O Type: _____

2) Predict the products and balance each equation. If no reaction occurs, write NR. *(16 pts)*

a. $Ca + Br_2 \rightarrow$ _____

b. $Cl_2 + MgI_2 \rightarrow$ _____

c. $Zn + NaCl \rightarrow$ _____

d. $C_7H_{16} + O_2 \rightarrow$ _____

e. $BaCl_2 \rightarrow$ _____

f. $NaBr(aq) + KNO_3(aq) \rightarrow$ _____

g. $Mg + Fe(NO_3)_3 \rightarrow$ _____

h. Lithium sulfate (aq) reacts with barium chloride (aq).

3) A student plans to decompose calcium bromide, $CaBr_2$, into its elements. *(6 pts)*

a. Write the balanced equation for the reaction.

b. Write the electron configuration for calcium. You may use full or noble gas notation.

c. The reaction produces 6.00 g of calcium metal. How many moles of calcium is this? Show your work.

d. Explain why one of the products is written as a diatomic molecule (Br_2) instead of Br.

4) Predict the products, balance the equation, and write the total ionic and net ionic equations. If no reaction occurs, write NR. Record the spectator ions. *(12 pts)*

a. $Na_3PO_4(aq) + CaCl_2(aq)$
   Balanced:

   Total ionic:

   Net ionic:

   Spectators:

b. $Al(s) + Pb(NO_3)_2(aq)$
   Balanced:

   Total ionic:

   Net ionic:

   Spectators:

c. Aqueous potassium hydroxide is mixed with aqueous copper(II) nitrate.
   Balanced:

   Total ionic:

   Net ionic:

   Spectators: